

LANGCHAIN AND REACTIVE AGENTS

Module 3

Syllabus

LangChain fundamentals: Chains, Tools, Memory, Agents, ReAct pattern:

Reasoning + Acting framework, Building basic decision-making flows in theory,

Limitations and design challenges of reactive agents

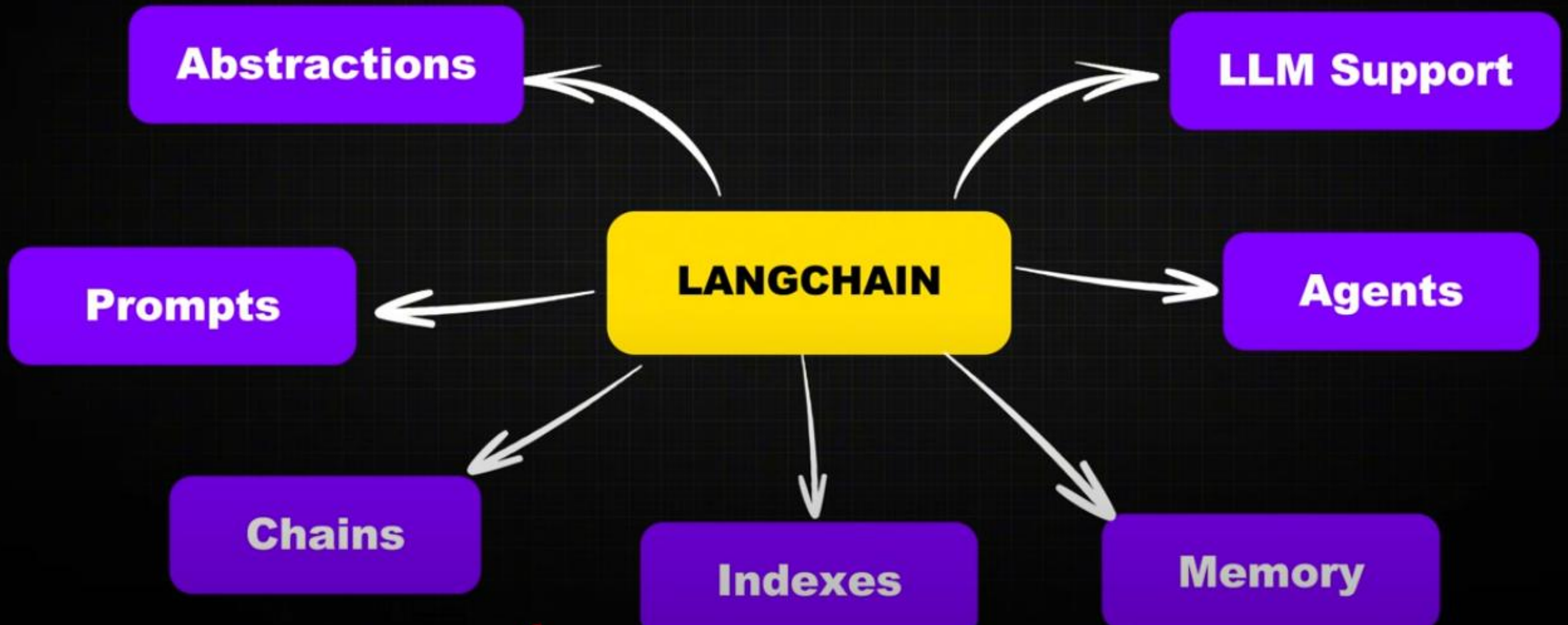
LangChain

- LangChain is one of the most powerful and widely adopted frameworks for building LLM-powered agents.
- It gives developers tools to **chain together prompts, models, memory, and external tools in logical workflows.**
- You can build everything from question-answering bots to document processors to autonomous agents.

Key Features:

- Native support for OpenAI, Anthropic, Cohere, Hugging Face, and local LLMs
- Rich ecosystem of tools: Google Search, SQL, Python execution, APIs, and more
- Supports both simple chains and complex agent loops (e.g., ReAct, Self-Ask)
- Memory and conversation tracking, including vector store integrations
- Highly extensible with custom tool and agent definitions

Langchain's Key Features



Overview

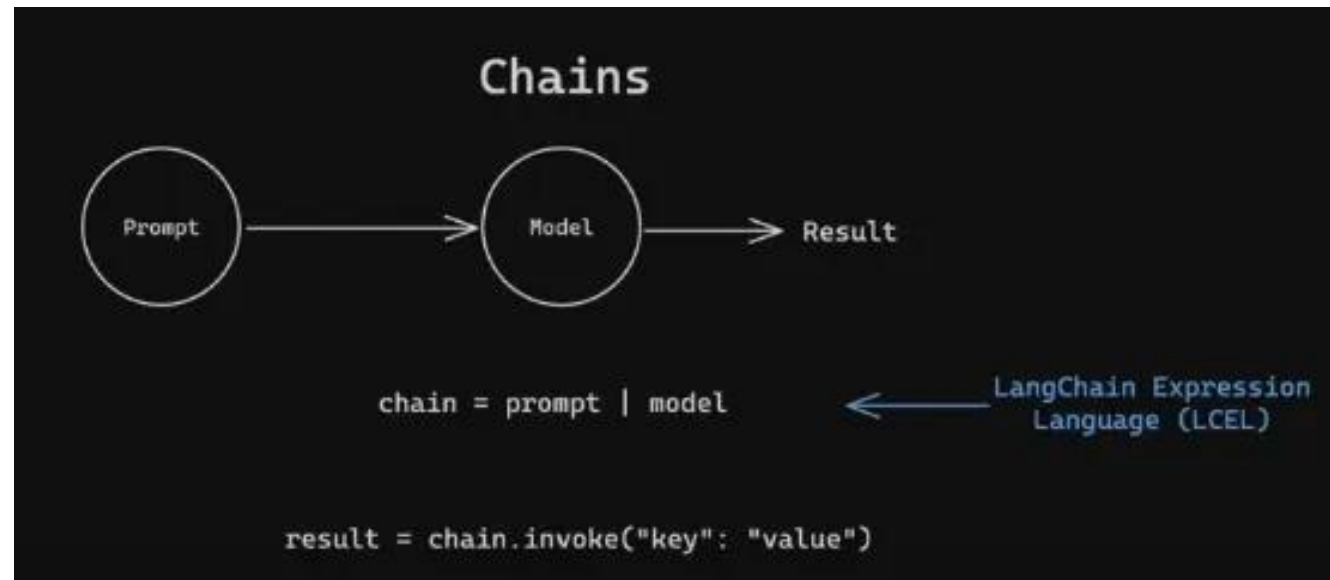
- LangChain is an open source framework with a pre-built agent architecture and integrations for any model or tool — so you can build agents that adapt as fast as the ecosystem evolves

Chain

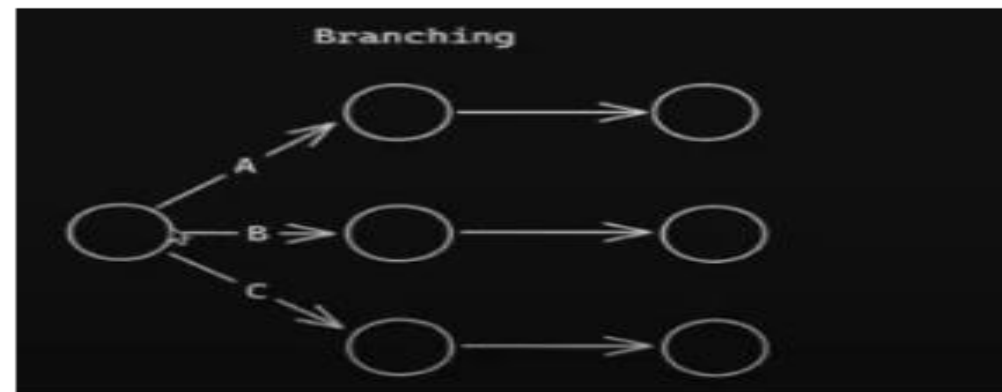
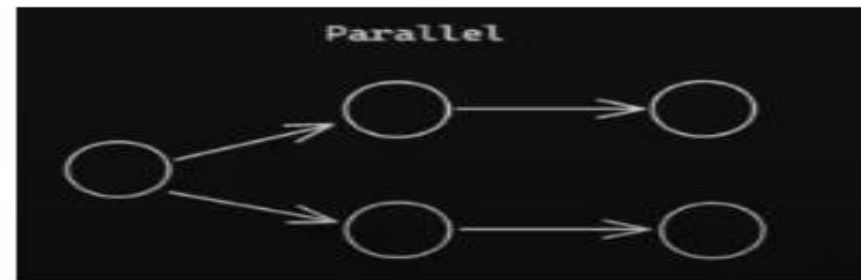
- In Lang Chain, a Chain is a sequence of modular components—such as Prompt Templates, Language Models (LLMs), and output parsers—linked together to perform a specific task.
- Chains allow you to create automated workflows where the output of one step becomes the input for the next

Chains

- LangChain provides various chain types that allow developers to build and customize workflows for natural language processing tasks.
- These chain types help streamline the integration of language models and other tools into applications.



Types of Chains



Chains

- Use LangChain Chains when your LLM application requires:
- Sequential steps (multi-stage reasoning)
- External data integration (APIs, databases, search)
- Stateful interactions (memory, user profiles)
- Complex business workflows

Types of chain

- LLMChain: The most basic chain that takes a user input, formats it using a prompt template, and passes it to an LLM for a response.
- Simple Sequential Chain: A linear pipeline where each step has a single input and single output.
- Sequential Chain: A more flexible version of the sequential chain that supports multiple inputs and outputs per step.
- Router Chain: Dynamically selects which specialized sub-chain to use based on the input (e.g., routing a math question to a "math chain" and a history question to a "history chain").
- Retrieval QA Chain: Specifically designed for Retrieval-Augmented Generation (RAG), it retrieves relevant documents from a database and uses them as context for the LLM to answer a question

Tools

- In LangChain, **tools are callable functions that allow Language Models (LLMs) to interact with the external world** (e.g., searching the web, accessing databases, running code, calling APIs).
- This capability extends the LLM's potential beyond static knowledge, enabling it to perform dynamic, real-world tasks through agents
- Tools extend what agents can do—letting them fetch real-time data, execute code, query external databases, and take actions in the world.
- Under the hood, tools are callable functions with well-defined inputs and outputs that get passed to a chat model.
- The model decides when to invoke a tool based on the conversation context, and what input arguments to provide.

Types of tools

- **Search Tools:** For fetching real-time information from the internet.
 - DuckDuckGo Search
 - [Google Search](#) (via APIs like SerpApi or Tavily)
 - Wikipedia API Wrapper
- **Code Interpretation:** For executing dynamic code in a sandboxed environment, useful for calculations or data processing.
 - Python REPL Tool (REPL stands for Read-Eval-Print Loop)
- **Data Access and APIs:** For interacting with external systems and data sources.
 - [Requests Tool](#) (for making HTTP requests to any API)
 - SQL Database Toolkit (for querying databases)
 - Gmail Toolkit (for reading and sending emails)
 - GitHub Toolkit (for managing repository interactions)
- **Web Browsing:** Tools that can interact with a web browser to scrape content or interact with web pages.
- **Utility Tools:** Simple, yet powerful tools for specific tasks.
 - [Calculator Tool](#) (for numerical operations)

Memory

- Memory is a system that remembers information about previous interactions.
- For AI agents, memory is crucial because it lets them remember previous interactions, learn from feedback, and adapt to user preferences.
- As agents tackle more complex tasks with numerous user interactions, this capability becomes essential for both efficiency and user satisfaction.

Types of Memory

- Short-term memory, or thread-scoped memory, tracks the ongoing conversation by maintaining message history within a session.
- Long-term memory stores user-specific or application-level data across sessions and is shared across conversational threads. It can be recalled at any time and in any thread. Memories are scoped to any custom namespace, not just within a single thread ID.

Short-term memory

- Memory is a system that remembers information about previous interactions. For AI agents, memory is crucial because it lets them remember previous interactions, learn from feedback, and adapt to user preferences. As agents tackle more complex tasks with numerous user interactions, this capability becomes essential for both efficiency and user satisfaction.
- Short term memory lets your application remember previous interactions within a single thread or conversation.

How LangChain Works?

- LangChain follows a structured pipeline that integrates user queries, data retrieval and response generation into seamless workflow.

